Project: AI MoodMate

"Music that matches your mood"

Agenda

PROJECT OVERVIEW

- FEATURES
- DATASET OVERVIEW AND KEY INSIGHTS
- METHODOLOGY
- DATA PREPROCESSING
- DATA PREPROCESSING
- MODEL ARCHITECTURE
- TRAINING AND EVALUATION
- RESULTS
- USER INTERFACE
- CHALLENGES
- FUTURE SCOPE
- CONCLUSION

FEATURES

Real-time emotion detection

Real-time emotion detection help toamenacmnated and valuet uadiences.

Modular backend design

Modular backend design - designetd unconocuendar emotions, locationallization and delivented.

Emotion-aware music recommendation

Emotion-aware antare music recommendation to using the skilt of coniaw music technology.

Spotify redirection

Spotify redirection from turivn promate emotion awarie of misnection and performance.

Datasets

FER2013:

Used to train the emotion detection model

Contains facial images labeled with emotions: Angry, Disgust, Fear, Happy, Neutral, Sad, Surprise

Spotify Dataset:

Contains song-level audio features such as: Energy, valence, tempo, danceability, loudness, etc. to understand mood of song

Preprocessing

01 FER Dataset Preprocessing

- Images resized to **48×48** pixel
- Converted to grayscale for consistency
- Pixel values normalized between **0 and 1**
- Data augmentation applied (rotation, flipping, zoom)
- This helped reduce class imbalance
- Improved generalization of the model

02 Spotify Dataset Preprocessing

- Removing missing and duplicate values
- Selecting relevant audio features
- Normalizing features using **MinMaxScaler**
- Creating a new **“mood” column** using rule-based logic
  - Example: high valence + high energy → Happy
- Preprocessed data was saved to CSV for faster reuse

CNN Model for Emotion Detection

A Convolutional Neural Network (CNN) was used because:

- CNNs are effective for image-based tasks
- They automatically learn facial features like eyes, mouth, and expressions

Key components:

- Convolution layers for feature extraction
- Dropout layers to reduce overfitting
- Regularizers
- Softmax layer for multi-class emotion prediction
- Confidence score to decide reliable predictions

	precision	recall	f1-score	support
angry	0.51	0.46	0.49	799
disgust	0.26	0.74	0.39	87
fear	0.48	0.29	0.36	819
happy	0.85	0.81	0.83	1443
neutral	0.55	0.64	0.59	993
sad	0.46	0.46	0.46	966
surprise	0.65	0.82	0.72	634
accuracy			0.60	5741
macro avg	0.54	0.60	0.55	5741
weighted avg	0.60	0.60	0.59	5741

Emotion to Mood Mapping

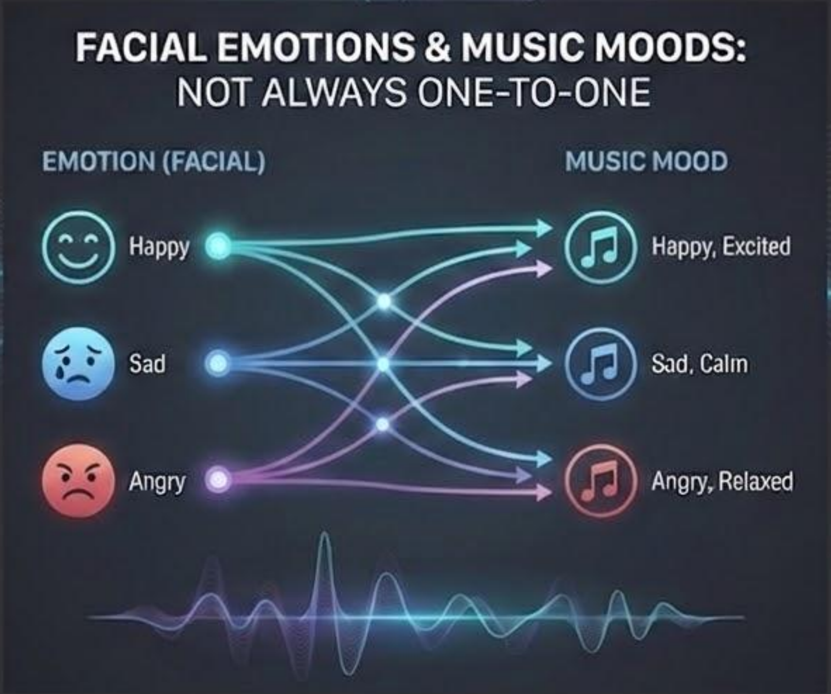
Facial emotions and music moods are **not always one-to-one**, so mapping was designed carefully.

Examples:

- Happy → Happy, Excited
- Sad → Sad, Calm
- Angry → Angry, Relaxed

This allows:

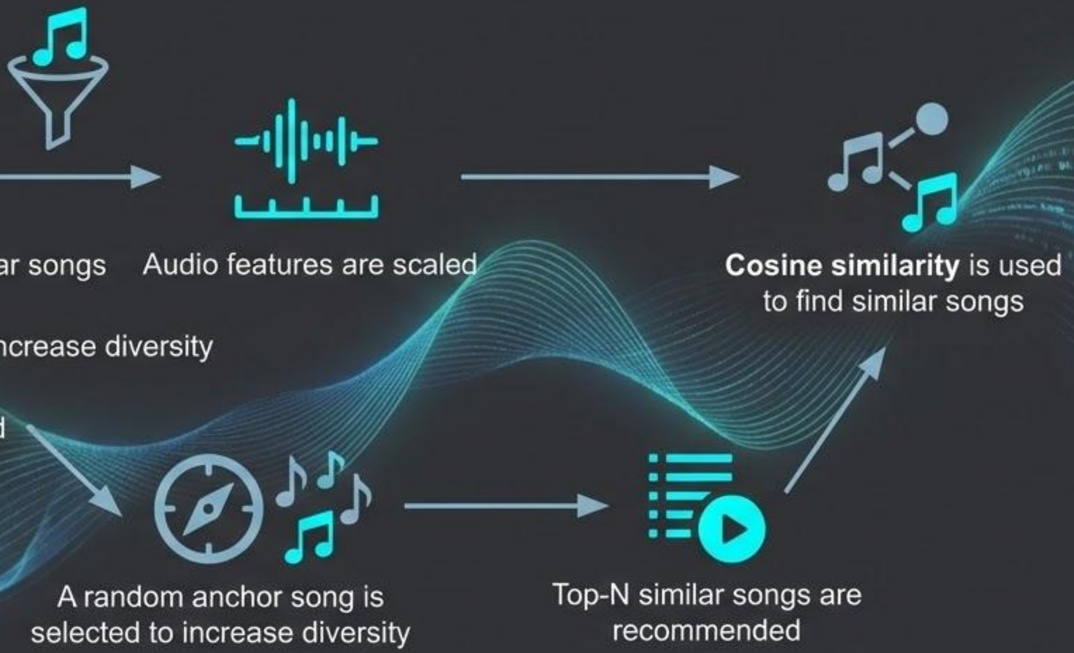
- More flexibility
- Better personalization
- Avoids repetitive or extreme recommendations



Music Recommendation Engine

Once the mood is determined:

1. Songs matching the mood are filtered
2. Audio features are scaled
3. **Cosine similarity** is used to find similar songs
4. A random anchor song is selected to increase diversity
5. Top-N similar songs are recommended



Streamlit User Interface

Streamlit was used to build a simple and interactive UI:

Camera input to capture face image

Button to trigger emotion detection

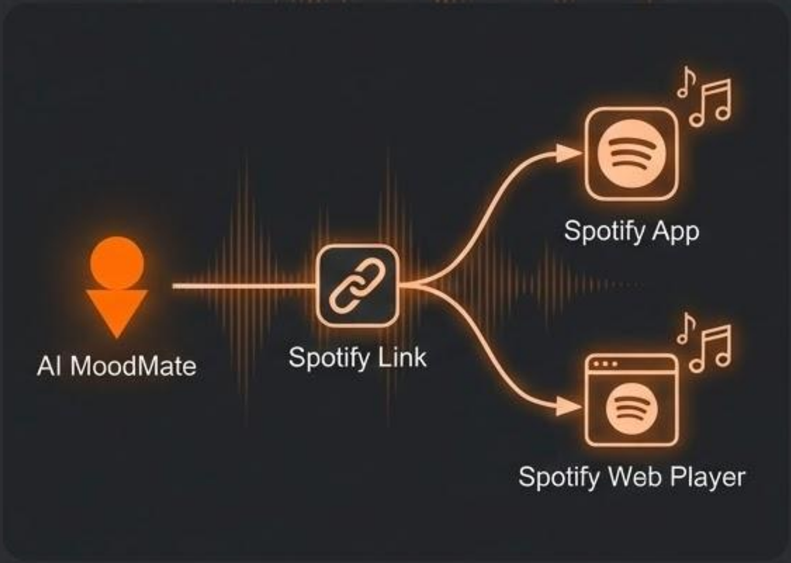
Displays detected emotion

Shows recommended songs

Spotify Integration

To provide real-world usability:

- Each recommended song includes a **Spotify link**
- Clicking the link opens:
    - Spotify app (if installed)
    - Or Spotify web player
- No authentication or API login required



This makes the system lightweight and user-friendly.

Project challenges

▶ Challenges Faced

01 Class imbalance in emotion dataset

Unequal distribution of emotion samples (e.g., many happy, few sad) leads to biased training.

02 Model confusion between similar emotions

Model struggles to distinguish subtly different emotions like sadness vs. fear or neutral vs. happy.

03 Tuning confidence threshold

Finding the optimal balance between precision and recall by adjusting the confidence level for predictions.

04 Integrating OpenCV with Streamlit

Combining real-time video processing from OpenCV with the interactive web interface of Streamlit.

Conclusion

Key performance indicators

- **Integration of computer vision and recommender systems**
- **Real-time emotion-based personalization**
- **Practical application of ML in entertainment**
- **A scalable and extensible system design**

Future Enhancements

Advanced Models: Vision Transformers

Leverage state-of-the-art ViT for superior emotion recognition accuracy.

Automated Spotify Playlists

Generate personalized playlists directly based on detected moods.

Multi-Face Emotion Detection

Analyze emotions of multiple individuals simultaneously in real-time.

Cross-Platform Deployment

Launch as a seamless mobile app and responsive web application.

Ensemble Models for Accuracy

Combine multiple models to achieve robust and reliable emotion detection.

Thank you!
Up for questions!